IADT Guidelines for the Management of Traumatic Dental Injuries

Part 2: Avulsion of Permanent Teeth & Part 3: Luxation Injuries — Clinical Decision Summary
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1. Overview

Traumatic dental injuries (TDIs) are time-critical emergencies in which prognosis is highly dependent on the elapsed time and conditions between injury and treatment. This summary condenses the International Association of Dental Traumatology (IADT) consensus guidelines for avulsion and luxation injuries of permanent teeth into a chairside decision aid.

2. Avulsion — First Aid Decision Tree

Step 1. Is replantation at the site of the accident possible?

- YES — Hold the tooth by the crown only (never touch the root). Rinse briefly with saline or the patient's saliva if visibly contaminated. Reposition into the socket immediately and have the patient bite on gauze. Seek dental care immediately.
- NO — Place the tooth in a physiologic storage medium immediately (see ranked list below) and transport the patient to a dental facility without delay. Do NOT store dry and do NOT scrub the root surface.

3. Storage Media — Ranked by Periodontal Ligament (PDL) Cell Preservation

Rank	Medium	Approx. PDL cell viability window	Notes
1	Hank's Balanced Salt Solution (HBSS)	Up to 24 hours	Gold standard; commercially available as “Save-A-Tooth”
2	Cold milk (preferably low-fat/skim)	3–6 hours	Most practical, widely available; physiologic osmolality/pH
3	Saliva (buccal vestibule)	Short term only	Low certainty evidence; use only if no better medium avail
4	Normal saline (0.9%)	Short term, suboptimal	Inferior to milk/HBSS; last-resort option
—	Tap water	NOT recommended	Hypotonic — causes PDL cell lysis; avoid

4. Extra-Oral Dry Time — Prognostic Categories

Dry Time	PDL Status	Clinical Implication
< 15 minutes, replanted at scene	Likely viable	Best prognosis; favor immediate replantation, flexible splint 2 weeks
< 60 minutes, stored in physiologic medium	Likely viable but compromised	Replant as soon as possible; systemic doxycycline/penicillin per protocol
≥ 60 minutes (regardless of storage medium)	Non-viable	Root resorption / ankylosis expected; different long-term management (e

Key principle: **replantation within 60 minutes** of avulsion, in an appropriate storage medium, is the single greatest determinant of favorable long-term prognosis. Do not delay transport to search for an ideal medium.

5. Post-Replantation Protocol (Permanent Teeth)

- Splint: flexible, passive splint for 2 weeks (extend to 4 weeks if there is an associated alveolar fracture).
- Antibiotics: systemic doxycycline (age/weight-appropriate dosing) is commonly recommended for 7 days in patients able to take tetracyclines; amoxicillin/penicillin V used as alternative in younger children due to tooth-staining risk with tetracyclines.
- Tetanus prophylaxis: verify status if the tooth contacted soil.
- Endodontic therapy: initiate root canal treatment within 7–10 days of replantation in teeth with closed apices, prior to splint removal.
- Follow-up: clinical and radiographic review at 2 weeks, 4 weeks, 3 months, 6 months, and 1 year, then annually.

6. Luxation Injuries — Classification and Management Summary

Injury Type	Clinical Presentation	Primary Management
Concussion	Tender to touch/percussion; no displacement; no mobility	No treatment needed; monitor pulp status
Subluxation	Tender, increased mobility, no displacement; may have sulcular bleeding	Monitor; flexible splint up to 2 weeks only if uncomfortable for patient
Extrusive luxation	Tooth appears elongated; excessive mobility	Reposition and stabilize with flexible splint 2 weeks; monitor pulp
Lateral luxation	Tooth displaced, usually palatally/lingually; often with alveolar bone fracture; may be immobile ("locked")	Reposition; flexible splint 4 weeks; monitor pulp; may need endodontic
Intrusive luxation	Tooth displaced axially into socket; appears shortened; may be immobile	Depends on root maturity: allow spontaneous re-eruption (open apex,

7. Special Consideration — Primary (Deciduous) Teeth

Avulsed primary teeth are **not replanted** due to the risk of damage to the developing permanent tooth bud. Reassure caregivers, check for fragment retention or intrusion into the permanent tooth follicle radiographically, and refer for routine follow-up.

References: Fouad AF, Abbott PV, Tsilingaridis G, et al. International Association of Dental Traumatology guidelines for the management of traumatic dental injuries: 2. Avulsion of permanent teeth. Dent Traumatol. 2020;36:331-342. — Bourguignon C, Cohenca N, Lauridsen E, et al. International Association of Dental Traumatology guidelines for the management of traumatic dental injuries: 1. Fractures and luxations. Dent Traumatol. 2020;36:314-330.